Bank Marketing

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MIS450-1: Data Mining

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Bank Marketing

Summary Statistics

The Bank Marketing dataset contains client information, campaign-related contact data, and economic indicators. The dataset includes a mixture of numeric (Table 1 & Figure 1) and categorical variables (Table 2).

From the categorical variables, the dataset is heavily concentrated in a few groups. Most clients (79.02%) were contacted in May - August, suggesting the marketing campaign was seasonal. In addition, the most common job categories indicate the sample includes a large portion of working professionals and skilled labor clients. Most clients were married (60.91%), which may influence financial decision-making and term deposit subscription behavior.

The dataset also includes several variables with “unknown” values, which represent missing information. For example, the default variable contains a notable percentage of unknown (19.50%), while housing and loan contain smaller “unknown” groups (2.55% each). These missing categories may require careful treatment during modeling, either by keeping “unknown” as a valid category or applying preprocessing techniques.

For the numeric variables, the average client age is approximately 40 years, suggesting the dataset is primarily composed of working-age clients. Several campaign variables show strong skewness. For example, the average number of contacts during the current campaign is low, but the maximum is much higher, indicating a small number of clients were contacted repeatedly. Similarly, pdays shows a median of 999, which reflects that many clients were not previously contacted in earlier campaigns. Several numeric variables show potential outliers, particularly duration and campaign, where the maximum values are substantially larger than the mean and standard deviation. In contrast, the economic indicators (euribor3m, nr.employed, and price/confidence indexes) show relatively stable ranges with less extreme variation.

The target variable (y) is highly imbalanced, with most observations labeled “no” (89.05%). This imbalance is important because a model could achieve high overall accuracy by predicting “no” for most clients while still failing to correctly identify clients who subscribe. Therefore, model performance should be evaluated using metrics that emphasize the minority class, such as recall/sensitivity for the “yes” outcome.

Table 1

Summary Statistics for Numeric Variables

The current time is 15:06:48 and the date is 09FEB2026

Variable	Mean	Std Dev	Minimum	Maximum	Median	N
age	40.1136198	10.3133615	18.0000000	88.0000000	38.0000000	4119
duration	256.7880554	254.7037361	0	3643.00	181.0000000	4119
campaign	2.5372663	2.5681592	1.0000000	35.0000000	2.0000000	4119
pdays	960.4221899	191.9227858	0	999.0000000	999.0000000	4119
previous	0.1903375	0.5417883	0	6.0000000	0	4119
emp.var.rate	0.0849721	1.5631145	-3.4000000	1.4000000	1.1000000	4119
cons.price.idx	93.5797043	0.5793488	92.2010000	94.7670000	93.7490000	4119
cons.conf.idx	-40.4991017	4.5945775	-50.8000000	-26.9000000	-41.8000000	4119
euribor3m	3.6213557	1.7335912	0.6350000	5.0450000	4.8570000	4119
nr.employed	5166.48	73.6679036	4963.60	5228.10	5191.00	4119

Note. This table provides the summary statistics for the numeric variables in the bank dataset.

Table 2

Summary Statistics for Categorical Variables

The current time is 15:31:55 and the date is 09FEB2026

month	Frequency	Percent
apr	215	5.22
aug	636	15.44
dec	22	0.53
jul	711	17.26
jun	530	12.87
mar	48	1.17
may	1378	33.45
nov	446	10.83
oct	69	1.68
sep	64	1.55

education	Frequency	Percent
basic.4y	429	10.42
basic.6y	228	5.54
basic.9y	574	13.94
high.school	921	22.36
illiterate	1	0.02
professional.course	535	12.99
university.degree	1264	30.69
unknown	167	4.05

marital	Frequency	Percent
divorced	446	10.83
married	2509	60.91
single	1153	27.99
unknown	11	0.27

job	Frequency	Percent
admin.	1012	24.57
blue-collar	884	21.46
entrepreneur	148	3.59
housemaid	110	2.67
management	324	7.87
retired	166	4.03
self-employed	159	3.86
services	393	9.54
student	82	1.99
technician	691	16.78
unemployed	111	2.69
unknown	39	0.95

day_of_week	Frequency	Percent
fri	768	18.65
mon	855	20.76
thu	860	20.88
tue	841	20.42
wed	795	19.30

default	Frequency	Percent
no	3315	80.48
unknown	803	19.50
yes	1	0.02

housing	Frequency	Percent
no	1839	44.65
unknown	105	2.55
yes	2175	52.80

loan	Frequency	Percent
no	3349	81.31
unknown	105	2.55
yes	665	16.14

y	Frequency	Percent
no	3668	89.05
ye	451	10.95

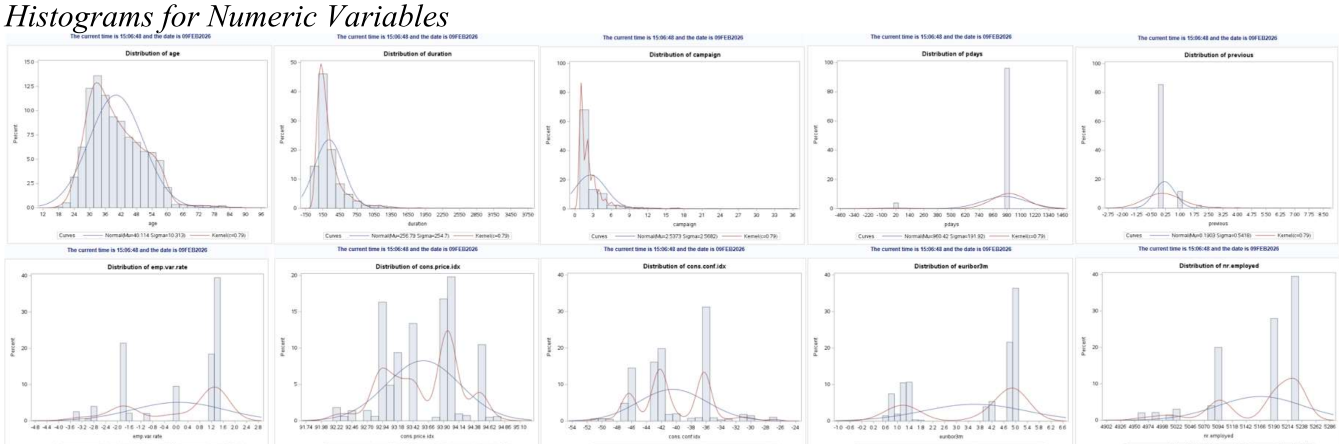
contact	Frequency	Percent
cellular	2652	64.38
telephone	1467	35.62

poutcome	Frequency	Percent
failure	454	11.02
nonexistent	3523	85.53
success	142	3.45

Note. This table provides the summary statistics for the categorical variables in the bank dataset.

Figure 1

Histograms for Numeric Variables



Note. This figure shows the histograms for the numeric variables in the bank dataset.

Anomaly Detection

Because several potential outliers were identified during the summary statistics analysis, a z-score based anomaly detection method was used to further examine extreme observations in the dataset. Z-scores standardize each numeric variable by measuring how far a value deviates from the mean in terms of standard deviations (Goss, 2022). This approach is useful for identifying unusually large or small observations across variables with different scales. In this analysis, observations with standardized values greater than 3 or less than −3 were flagged as anomalies. The SAS code used to populate this output is in Appendix A.

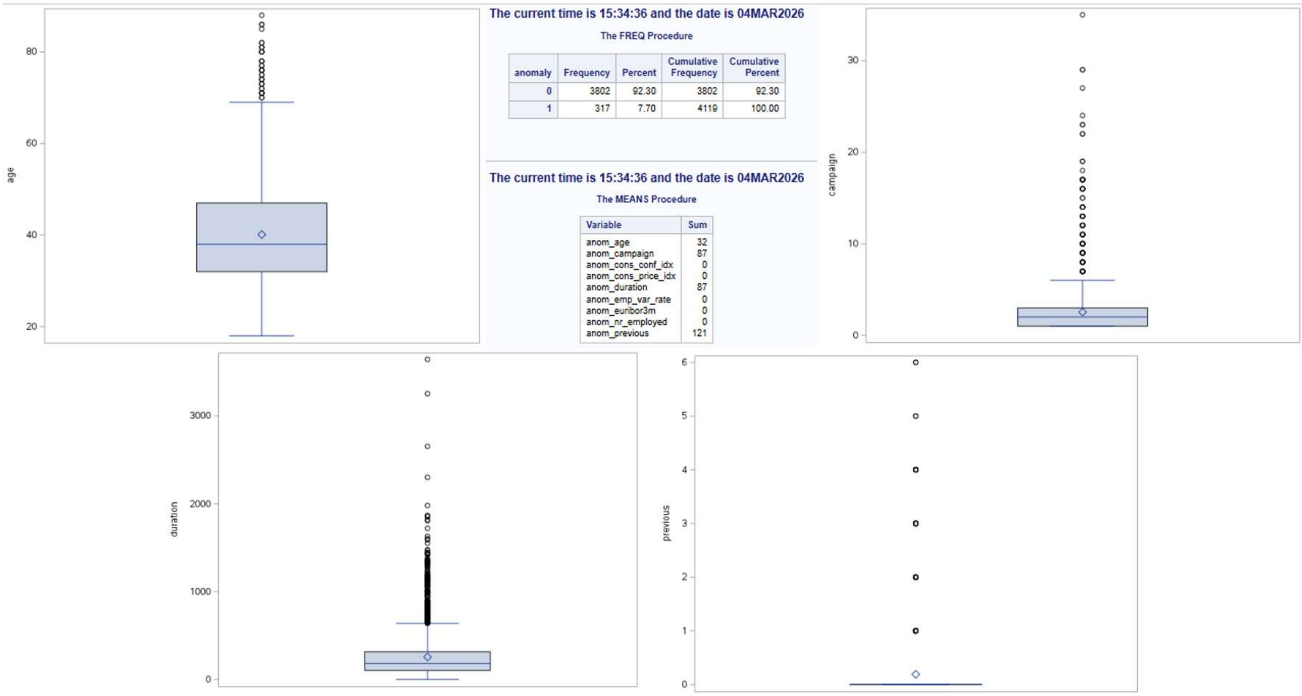
As shown in Figure 2, The anomaly analysis identified 317 observations (7.7% of the dataset). While this proportion is relatively small, it indicates that several variables contain observations that differ substantially from the majority of the data. These cases helps determine whether extreme values are the result of data errors or reflect meaningful variations in customer behavior.

Among the numeric variables examined, ‘previous’ contained the largest number of anomalies (121 observations). This variable represents the number of contacts performed before the current marketing campaign. The high number of anomalies likely reflects the distribution of the data, where most customers had little to no previous contact with the bank, while a smaller number had been contacted many times before. This uneven distribution can produce extreme standardized values. Similarly, the variables campaign and duration each contained 87 anomalous observations, indicating that a subset of customers required significantly more contact attempts or experienced longer call durations than the typical observation.

The anomalies appear to reflect natural variation within the marketing campaign data rather than clear data entry errors. For example, some customers may require multiple contact attempts before making a decision, or conversations may vary greatly in length depending on the complexity of the interaction. Because these values may contain useful behavioral information, the observations were retained in the dataset for subsequent analysis and predictive modeling.

Figure 2

Summary Results from Z-Score Anomaly Detection and Boxplots of Selected Variables



The current time is 15:34:36 and the date is 04MAR2026

The FREQ Procedure

anomaly	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	3802	92.30	3802	92.30
1	317	7.70	4119	100.00

The current time is 15:34:36 and the date is 04MAR2026

The MEANS Procedure

Variable	Sum
anom_age	32
anom_campaign	87
anom_cons_conf_idx	0
anom_cons_price_idx	0
anom_duration	87
anom_emp_var_rate	0
anom_euribor3m	0
anom_nr_employed	0
anom_previous	121

Note. This figure presents boxplots and summary output generated for selected numeric variables in the bank marketing dataset. The boxplots display the distribution of the variables and highlight potential extreme observations. The accompanying tables summarize the frequency of observations flagged during the anomaly detection procedure.

Association Among Variables

To examine associations among the quantitative variables, Pearson correlation analysis was conducted. Pearson's correlation coefficient measures the strength and direction of the linear relationship between two continuous variables, making it an appropriate method for evaluating relationships among numeric predictors such as age, duration, campaign contacts, and economic indicators. Because many of the variables in the bank marketing dataset are measured on continuous scales, Pearson correlation provides a straightforward way to identify whether increases or decreases in one variable tend to correspond with changes in another. Identifying these relationships is an important step in exploratory data analysis because it helps reveal patterns within the data and can highlight potential multicollinearity among predictors before building predictive models (Schober et al., 2018). In addition to the correlation matrix, a scatterplot matrix (Figure 4) was generated to visually examine pairwise relationships among the quantitative variables and confirm patterns identified in the correlation analysis.

The results of the correlation analysis (Figure 3) reveal several meaningful relationships among the quantitative variables. Euribor3m shows a very strong positive correlation with emp_var_rate ($r = 0.9703$) and nr_employed ($r = 0.9426$). Emp_var_rate and nr_employed also demonstrate a strong positive relationship ($r = 0.8972$). These high correlation values suggest that the variables move closely together and likely reflect similar underlying conditions. Additional moderate relationships are observed between cons_price_idx and emp_var_rate ($r = 0.7552$), as well as between cons_price_idx and euribor3m ($r = 0.6576$), indicating that price index measures also tend to vary with broader indicators.

In contrast, the variables related to customer contact behavior, such as campaign, pdays, previous, and duration, generally exhibit weaker correlations with one another and with the economic indicators. This suggests that customer interaction patterns vary more independently across observations. Overall, the correlation analysis indicates that while economic variables demonstrate strong associations with one another, most customer-level variables show relatively weak linear relationships, highlighting the diversity of customer interaction patterns within the dataset.

Figure 3

Pearson Correlation Analysis of Quantitative Variables

```
data _null_;
call symput ('timenow',put (time(),time.));
call symput ('datenow',put (date(),date9.));
run;

title "The current time is &timenow and the date is &datenow";

proc corr data=MIS450.bank;
var age duration campaign pdays previous euribor3m emp_var_rate
    nr_employed cons_price_idx cons_conf_idx;
run;
```

The current time is 18:10:51 and the date is 04MAR2026

The CORR Procedure

10 Variables:	age duration campaign pdays previous euribor3m emp_var_rate nr_employed cons_price_idx cons_conf_idx

Simple Statistics

Variable	N	Mean	Std Dev	Sum	Minimum	Maximum
age	4119	40.11362	10.31336	165228	18.00000	88.00000
duration	4119	256.78806	254.70374	1057710	0	3643
campaign	4119	2.53727	2.56816	10451	1.00000	35.00000
pdays	4119	960.42219	191.92279	3955979	0	999.00000
previous	4119	0.19034	0.54179	784.00000	0	6.00000
euribor3m	4119	3.62136	1.73359	14916	0.63500	5.04500
emp_var_rate	4119	0.08497	1.56311	350.00000	-3.40000	1.40000
nr_employed	4119	5166	73.66790	21280738	4964	5228
cons_price_idx	4119	93.57970	0.57935	385455	92.20100	94.76700
cons_conf_idx	4119	-40.49910	4.59458	-166816	-50.80000	-26.90000

Pearson Correlation Coefficients, N = 4119
Prob > |r| under H0: Rho=0

	age	duration	campaign	pdays	previous	euribor3m	emp_var_rate	nr_employed	cons_price_idx	cons_conf_idx
age	1.00000	0.04130 0.0080	-0.01417 0.3633	-0.04342 0.0053	0.05093 0.0011	-0.01503 0.3348	-0.01919 0.2182	-0.04194 0.0071	-0.00048 0.9753	0.09814 <.0001
duration	0.04130 0.0080	1.00000	-0.08535 <.0001	-0.04700 0.0026	0.02572 0.0988	-0.03233 0.0380	-0.02885 0.0641	-0.04422 0.0045	0.01667 0.2847	-0.03474 0.0258
campaign	-0.01417 0.3633	-0.08535 <.0001	1.00000	0.05874 0.0002	-0.09149 <.0001	0.15944 <.0001	0.17608 <.0001	0.16104 <.0001	0.14502 <.0001	0.00788 0.6130
pdays	-0.04342 0.0053	-0.04700 0.0026	0.05874 0.0002	1.00000	-0.58794 <.0001	0.30148 <.0001	0.27068 <.0001	0.38198 <.0001	0.05847 0.0002	-0.09209 <.0001
previous	0.05093 0.0011	0.02572 0.0988	-0.09149 <.0001	-0.58794 <.0001	1.00000	-0.45885 <.0001	-0.41524 <.0001	-0.51485 <.0001	-0.16492 <.0001	-0.05142 0.0010
euribor3m	-0.01503 0.3348	-0.03233 0.0380	0.15944 <.0001	0.30148 <.0001	-0.45885 <.0001	1.00000	0.97031 <.0001	0.94259 <.0001	0.65716 <.0001	0.27660 <.0001
emp_var_rate	-0.01919 0.2182	-0.02885 0.0641	0.17608 <.0001	0.27068 <.0001	-0.41524 <.0001	0.97031 <.0001	1.00000	0.89717 <.0001	0.75515 <.0001	0.19502 <.0001
nr_employed	-0.04194 0.0071	-0.04422 0.0045	0.16104 <.0001	0.38198 <.0001	-0.51485 <.0001	0.94259 <.0001	0.89717 <.0001	1.00000	0.47256 <.0001	0.10705 <.0001
cons_price_idx	-0.00048 0.9753	0.01667 0.2847	0.14502 <.0001	0.05847 0.0002	-0.16492 <.0001	0.65716 <.0001	0.75515 <.0001	0.47256 <.0001	1.00000	0.04583 0.0033
cons_conf_idx	0.09814 <.0001	-0.03474 0.0258	0.00788 0.6130	-0.09209 <.0001	-0.05142 0.0010	0.27660 <.0001	0.19502 <.0001	0.10705 <.0001	0.04583 0.0033	1.00000

Note. Pearson correlation coefficients were calculated using PROC CORR to measure the strength and direction of linear relationships among the quantitative variables. Values closer to

+-1 indicate stronger positive or negative relationships, while values near 0 indicate weak or no linear association (Schober et al., 2018). P-values shown beneath each coefficient indicate the statistical significance of the correlation.

Figure 4

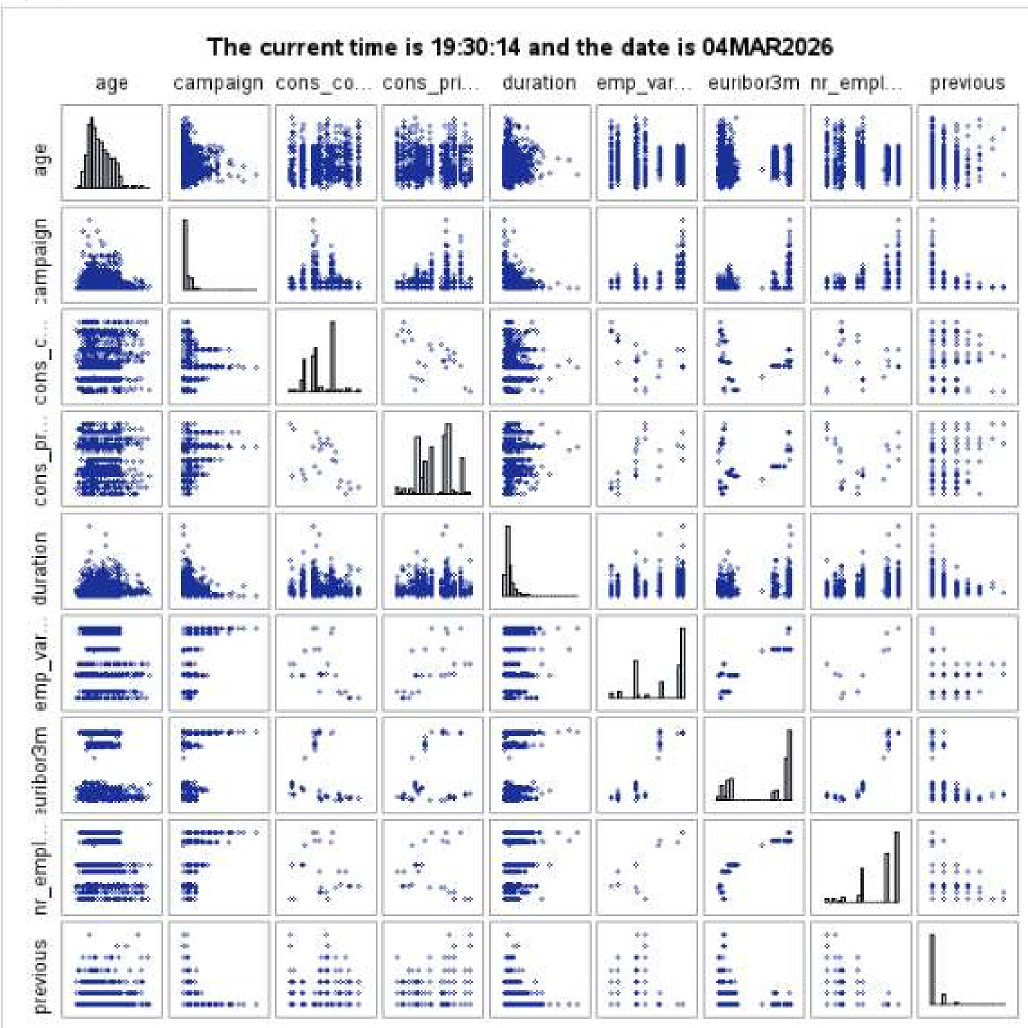
Scatterplot Matrix for the Quantitative Variables

```
data _null_;
call symput ('timenow',put (time(),time.));
call symput ('datenow',put (date(),date9.));
run;

title "The current time is &timenow and the date is &datenow";

proc sgscatter data=MIS450.BANK;
    matrix age campaign cons_conf_idx cons_price_idx duration emp_var_rate euribor3m nr_employed previous /
        diagonal=(histogram);

run;
```



Note. The scatterplot matrix displays pairwise relationships among the quantitative variables. Histograms along the diagonal illustrate the distribution of each variable. This visualization supports the Pearson correlation analysis by providing a visual representation of potential linear relationships and overall distribution patterns within the dataset.

Clustering

"The K-means partitional-clustering algorithm is the simplest and most commonly used algorithm employing a square-error criterion" (Kantardzic, 2019, p. 310). K-means method that divides observations into a predetermined number of clusters while minimizing the total within-cluster variation. The algorithm works by representing each cluster with a centroid and assigning observations to the cluster whose centroid is most similar, typically based on Euclidean distance. Because this method focuses on minimizing within-cluster variation, observations grouped within the same cluster tend to share similar characteristics.

Compared with other clustering techniques (hierarchical, density-based, or distribution-based), K-means is computationally efficient and well suited for datasets containing a moderate number of observations with primarily numerical variables. Partition-based clustering algorithms are widely used in data mining due to their relatively low computational complexity and ability to scale to larger datasets (Xu & Tian, 2015). Because the bank marketing dataset contains over 4,000 observations and multiple quantitative variables, K-means provides an effective and interpretable method for identifying natural groupings within the data.

To determine the appropriate number of clusters, several clustering solutions were evaluated using values of k ranging from two to five clusters. Comparing multiple cluster solutions helps ensure that the selected model provides meaningful groupings without over-segmenting the data. The four-cluster solution was ultimately selected because it provided a strong improvement in cluster separation compared to the two- and three-cluster solutions, while maintaining reasonably interpretable cluster sizes. Although the five-cluster solution produced a slightly higher overall r-squared value, it resulted in very small clusters that offered limited interpretive value.

The four-cluster solution identified several distinct groupings within the dataset based on the quantitative variables. The cluster summary indicates that the majority of observations fall into Cluster 4 (n = 3,096), suggesting that most customers share relatively similar characteristics. Smaller segments were identified in Cluster 3 (n = 753), Cluster 2 (n = 158), and Cluster 1 (n = 112), indicating more specialized customer groups within the data. The overall r-squared value of approximately 0.81 suggests that the clustering model explains a substantial portion of the variation in the dataset, indicating meaningful separation between clusters.

Examining the cluster means reveals that several variables contribute strongly to differentiating the clusters. In particular, duration, pdays, and previous interactions show notable differences across clusters. Cluster 1 is characterized by substantially higher call durations compared to the other clusters, indicating customers who spent more time on calls during the marketing campaign. Cluster 3 shows elevated values for the campaign variable, suggesting that customers in this group were contacted more frequently during the campaign period. In contrast, Cluster 4 represents the largest and most typical customer segment, with values near the overall averages for most variables.

Figure 5

FASTCLUS (k=4) Clustering of Quantitative Variables

```
data _null_;
call symput ('timenow',put (time(),time.));
call symput ('datenow',put (date(),date9.));
run;

title "The current time is &timenow and the date is &datenow";

proc fastclus data=MIS450.bank maxclusters=4 maxiter=100;
var age duration campaign pdays previous euribor3m emp_var_rate
    nr_employed cons_price_idx cons_conf_idx;
run;
```

The current time is 19:46:27 and the date is 04MAR2026

The FASTCLUS Procedure
Replace=FULL Radius=0 Maxclusters=4 Maxiter=100 Converge=0.02

Cluster Summary

Cluster	Frequency	RMS Std Deviation	Maximum Distance from Seed to Observation	Radius Exceeded	Nearest Cluster	Distance Between Cluster Centroids
1	112	138.4	2367.4		3	752.8
2	158	67.4000	810.6		4	1014.3
3	753	51.6396	385.0		4	371.1
4	3096	34.3151	283.2		3	371.1

Statistics for Variables

Variable	Total STD	Within STD	R-Square	RSQ/(1-RSQ)
age	10.31336	10.30209	0.002911	0.002919
duration	254.70374	123.98759	0.763207	3.223094
campaign	2.56816	2.56078	0.006464	0.006506
pdays	191.92279	21.66525	0.987266	77.531184
previous	0.54179	0.43842	0.345655	0.528247
euribor3m	1.73359	1.65428	0.090065	0.098979
emp_var_rate	1.56311	1.50601	0.072407	0.078059
nr_employed	73.66790	68.19270	0.143746	0.167878
cons_price_idx	0.57935	0.57852	0.003597	0.003610
cons_conf_idx	4.59458	4.57272	0.010215	0.010321
OVER-ALL	103.57387	45.42211	0.807816	4.203341

Pseudo F Statistic =	5765.58

Approximate Expected Over-All R-Squared =	0.72066

Cubic Clustering Criterion =	35.503

WARNING: The two values above are invalid for correlated variables.

Cluster Means

Cluster	age	duration	campaign	pdays	previous	euribor3m	emp_var_rate	nr_employed	cons_price_idx	cons_conf_idx
1	42.000000	1276.419643	2.455357	981.303571	0.107143	3.683125	0.082143	5168.660714	93.594170	-40.914286
2	42.253165	298.272152	1.784810	5.835443	1.784810	1.023411	-2.015823	5026.917089	93.408399	-38.372152
3	40.253652	523.876494	2.276228	999.000000	0.135458	3.649481	0.119124	5168.748738	93.598691	-40.950066
4	39.902132	152.824612	2.642119	999.000000	0.125323	3.744863	0.183979	5172.973966	93.583306	-40.482946

Cluster Standard Deviations

Cluster	age	duration	campaign	pdays	previous	euribor3m	emp_var_rate	nr_employed	cons_price_idx	cons_conf_idx
1	10.7535873	411.1562610	1.7649663	131.8303468	0.3384422	1.6839741	1.5554157	71.0233388	0.5713768	4.6197355
2	14.4948648	205.1041571	1.1305924	3.9225962	1.1135630	0.6627353	0.8554451	55.5823647	0.7962969	6.3742903
3	10.2495818	146.9599449	2.1214910	0.0000000	0.4096302	1.7135369	1.5397243	70.2460605	0.5646789	4.4020218
4	10.0389443	83.6357549	2.7292153	0.0000000	0.3844198	1.6736663	1.5217163	68.1631865	0.5688863	4.5017792

Note. Output from the SAS PROC FASTCLUS procedure showing the cluster summary, variable clustering statistics, and cluster profiles for the four-cluster model. The tables report cluster sizes, measures of cluster separation, and the mean and standard deviation of each quantitative variable within each cluster.

Classification Technique

The appropriate classification technique for this dataset is a decision tree because the target variable (y) is binary. A decision tree classifies observations by splitting the dataset into smaller subsets based on predictor variables that best separate the classes. This process creates a tree-like structure of decision rules that ultimately leads to a predicted outcome. Decision trees are considered logic-based classification models that divide the data into regions associated with specific class labels (Kantardzic, 2019).

Decision trees are also appropriate for this dataset because it contains many predictors, including both numeric and categorical variables. Unlike some machine learning techniques, decision trees can work with both types of variables without requiring extensive preprocessing (Kantardzic, 2019). The algorithm evaluates potential splits in the data and selects the predictors that best distinguish between clients who subscribe to the term deposit and those who do not.

Another reason a decision tree is useful in this situation is that the dataset contains a much higher number of “no” responses compared to “yes” responses. Because of this imbalance, it is important for the model to identify patterns that help separate the smaller group of clients who subscribe from the larger group who do not. The structure of a decision tree can also be easily interpreted, which makes it helpful for understanding which factors influence customer responses to marketing campaigns. For this analysis, a decision tree model was implemented using the PROC HPSPLIT procedure as shown in Figure 6.

The decision tree model identified several variables that play an important role in predicting whether a client subscribes to a term deposit. Based on the results, call duration was the most influential predictor in the model. This suggests that longer interactions with clients may increase the likelihood of a successful subscription. Other important variables included the number of employees in the economy (nr_employed), contact type, job category, and the number of days since a previous campaign (pdays). These results indicate that both characteristics of the marketing interaction and broader economic conditions contribute to predicting customer responses. The decision tree structure highlights how these variables interact to separate clients who subscribe from those who do not.

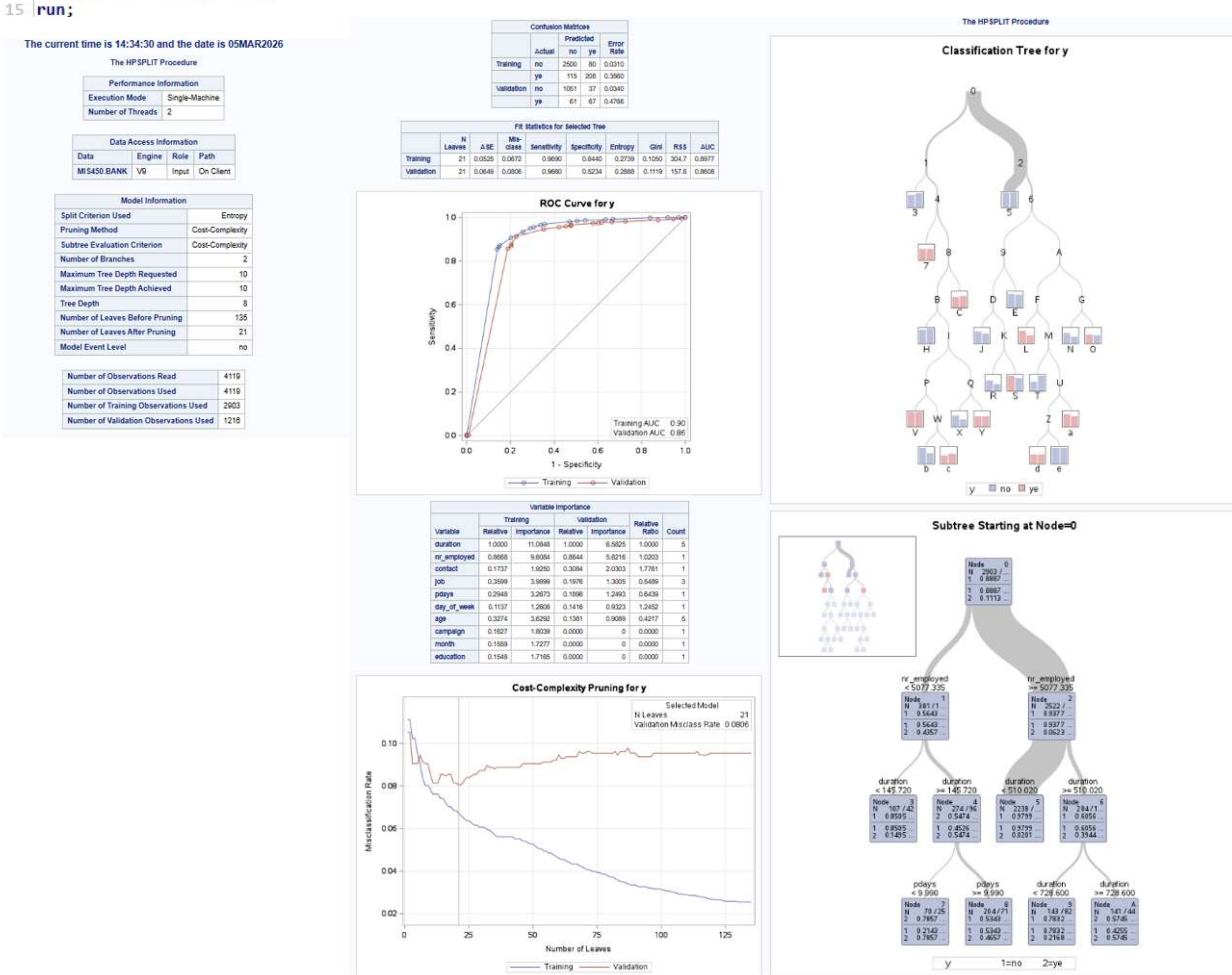
Figure 6

Decision Tree Classification Model for Predicting Client Subscription

```
data _null_;
call symput ('timenow',put (time(),time.));
call symput ('datenow',put (date(),date9.));
run;

title "The current time is &timenow and the date is &datenow";

proc hpsplit data=MIS450.BANK seed=1;
    class y job marital education default housing loan contact month day_of_week poutcome;
    model y =
        age duration campaign pdays previous emp_var_rate euribor3m nr_employed cons_price_idx cons_conf_idx
        job marital education default housing loan contact month day_of_week poutcome;
    partition fraction(validate=0.30);
    prune costcomplexity;
run;
```



Note. The model was developed using a decision tree classification technique implemented through PROC HPSPLIT. All categorical variables were specified in the CLASS statement, while both categorical and quantitative variables were included as predictors of the target variable y (client subscription outcome). A 30% validation partition was used to evaluate model performance, and cost-complexity pruning was applied to reduce overfitting. The seed value was set to ensure reproducibility of the model results.

Model Evaluation

The performance statistics indicate that the decision tree model fits the data well and is capable of accurately predicting whether a client will subscribe to a term deposit. The model achieved an area under the receiver operating characteristic curve (AUC) of approximately 0.90 for the training dataset and approximately 0.86 for the validation dataset. AUC values closer to 1.0 indicate stronger classification performance, and anything over 0.80 is generally considered strong (Corbacioglu & Aksel, 2023). This suggests that the model is highly effective at distinguishing between clients who subscribe and those who do not.

The confusion matrices and fit statistics further support the strength of the model. The validation misclassification rate was approximately 8%, meaning that the model correctly classified the outcome for the majority of observations. The relatively small difference between the training and validation AUC values suggests that the model generalizes well and does not appear to have significant overfitting. This is also supported by the use of cost-complexity pruning, which reduced the size of the tree to 21 leaves while maintaining strong predictive performance.

Although the decision tree model performs well, several improvements could be explored. One potential improvement would be further refining the decision tree by adjusting model complexity or applying additional pruning techniques. Another consideration would be evaluating the influence of dominant predictors (such as duration), which may strongly affect the model's predictions. Addressing the imbalance between "yes" and "no" responses could also improve the model's ability to correctly identify clients who subscribe. In addition to improving the decision tree itself, comparing the model with other classification techniques such as logistic regression or support vector machines (SVM) could provide insight into whether alternative approaches offer better predictive performance.

References

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Appendix A

SAS Code for Z-Score Standardization, Anomaly Detection, and Boxplot Visualization

```
data _null_;
call symput ('timenow',put (time(),time.));
call symput ('datenow',put (date(),date9.));
run;

title "The current time is &timenow and the date is &datenow";

proc standard data=MIS450.bank mean=0 std=1 out=bank_zscore;
var age campaign cons_conf_idx cons_price_idx duration
    emp_var_rate euribor3m nr_employed previous;
run;

data bank_anomalies;
  set bank_zscore;
  if max(of age campaign cons_conf_idx cons_price_idx duration
            emp_var_rate euribor3m nr_employed previous) > 3
     or
     min(of age campaign cons_conf_idx cons_price_idx duration
            emp_var_rate euribor3m nr_employed previous) < -3
  then anomaly = 1;
  else anomaly = 0;
run;

proc freq data=bank_anomalies;
  tables anomaly;
run;

data anomalies;
  set bank_z;

  anom_age            = (age > 3 or age < -3);
  anom_campaign       = (campaign > 3 or campaign < -3);
  anom_cons_conf_idx  = (cons_conf_idx > 3 or cons_conf_idx < -3);
  anom_cons_price_idx = (cons_price_idx > 3 or cons_price_idx < -3);
  anom_duration       = (duration > 3 or duration < -3);
  anom_emp_var_rate   = (emp_var_rate > 3 or emp_var_rate < -3);
  anom_euribor3m      = (euribor3m > 3 or euribor3m < -3);
  anom_nr_employed    = (nr_employed > 3 or nr_employed < -3);
  anom_previous       = (previous > 3 or previous < -3);
run;

proc means data=anom_flags sum maxdec=0;
  var anom_age anom_campaign anom_cons_conf_idx anom_cons_price_idx anom_duration
      anom_emp_var_rate anom_euribor3m anom_nr_employed anom_previous;
run;

proc sgplot data=MIS450.bank;
    vbox age;
run;

proc sgplot data=MIS450.bank;
    vbox campaign;
run;

proc sgplot data=MIS450.bank;
    vbox duration;
run;

proc sgplot data=MIS450.bank;
    vbox previous;
run;
```

Note. This SAS code standardizes the quantitative variables in the dataset using z-scores to identify potential anomalies. Observations with standardized values greater than 3 or less than −3 were flagged as anomalies. Additional indicator variables were created to determine which predictors contained extreme values, and summary statistics were generated to count anomalies for each variable. Boxplots were also produced to visually examine the distribution of selected variables and confirm the presence of extreme observations.